

# Likun Xie

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## Research Interests

Number theory: sieve methods and arithmetic over number fields and function fields; commutative algebra: homological methods and local cohomology.

## Appointments

**Max Planck Institute for Mathematics**, Bonn, Germany  
Postdoctoral Fellow.

July 2025–present

## Education

**University of Illinois Urbana–Champaign**, USA

2020–2025

Ph.D. in Mathematics. Advisors: Alexandru Zaharescu and Bruce C. Berndt.

**University of Warwick**, UK

2018–2019

M.Sc. in Mathematics, Distinction. Dissertation: *Descent, Affine Representability and Classifying Space in  $\mathbb{A}^1$ -Homotopy Theory*. Advisor: Marco Schlichting.

**University of Manchester**, UK

2016–2018

B.Sc. (Hons) in Mathematics and Physics, First Class. Dissertation: *Topos Foundation for Set Theory*. Advisor: Michael Prest.

## Publications and Preprints

- L. Xie, Products of Prime Ideals in Ray Class Groups (2026). arXiv:2606.30567
- L. Xie, Connectedness in Codimension One and the Non- $S_2$  Locus (2026). arXiv:2605.06617.
- L. Xie, On a conjecture of Erdős over function fields, *Mathematische Zeitschrift* **313** (2026), 64.
- L. Xie, Almost Prime Orders of Elliptic Curves Modulo Prime Powers (2025). arXiv:2504.18732.
- K. Nath and L. Xie, Almost primes and primes that are sums of two squares plus 1, *Acta Arithmetica* **223** (2026), 51–82.
- B. C. Berndt and L. Xie, Identities for the product of two Dirichlet series satisfying Hecke’s functional equation, *Journal of Mathematical Analysis and Applications* **550** (2025), no. 1, Paper No. 129514.
- L. Xie, On primes  $p$  such that  $p - b$  has a large prime power factor and few other divisors (2025). arXiv:2410.14133.
- L. Xie, On negative real moments of restricted factors in arithmetic progression, *The Ramanujan Journal* **67** (2025), 41.
- L. Xie, On an instance of the small Cohen–Macaulay conjecture, *Journal of Pure and Applied Algebra* **228** (2024), no. 3, 107491.
- B. C. Berndt, L. Xie, and A. Zaharescu, Proofs of McIntosh’s conjecture on Franel integrals and two generalizations, *Advances in Mathematics* **423** (2023), 109041.

## Awards

- Dr. Lois M. Lackner Mathematics Fellowship, University of Illinois Urbana–Champaign, 2023–2025.

## Teaching and Mentoring

- Mentored high-school and undergraduate interns in number theory projects involving elementary sieve methods at MPIM, 2025.
- Teaching Assistant, University of Illinois Urbana–Champaign: Calculus, Differential Equations, and Algebra, 2020–2023.

- Grader, University of Illinois Urbana–Champaign: Abstract Algebra, Honors Abstract Algebra, Commutative Algebra, Real Analysis, Complex Variables, and Differentiable Manifolds, 2020–2024.